A Survey and Benchmark Evaluation for Neural-Network-Based Lossless Universal Compressors Toward Multi-Source Data

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Problems & Ideas

- Problems: In recent years, due to the emergence of hardware acceleration devices such as GPUs and TPUs, the performance bottleneck of neural networks (NN) has been overcome, making NN-based compression algorithms increasingly practical and popular. However, the research survey for the NN-based universal lossless compressors has not been conducted yet, and there is also a lack of unified evaluation metrics.
- Ideas: To address the above problems, in this paper, we present a holistic survey as well as benchmark evaluations.

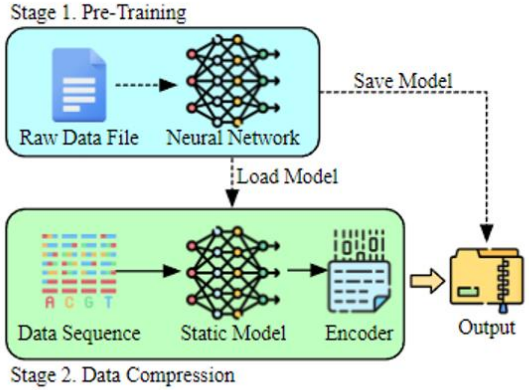


Fig. 1 The compression process of the static pre-training methods

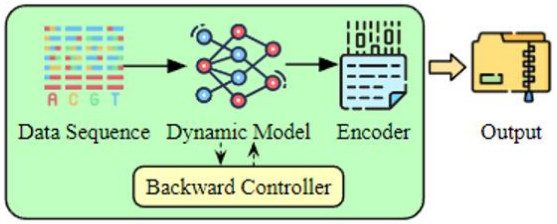


Fig. 2 Schematic of the compression process of the adaptive methods

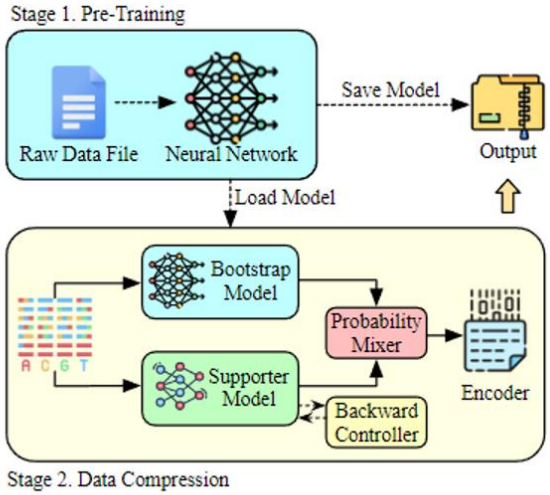


Fig. 3 The compression process of the semi-adaptive methods

Main Contributions

- Contributions:
  - We holistically reviewed NN-based universal compressors and unified the corresponding evaluation metrics.
  - We offered a comprehensive benchmark evaluation for NN-based compressors and conducted a thorough analysis of existing state-of-the-art solutions.
  - We identified the challenges of NN-based lossless compression and summarized potential research directions.

Table 1 Overall comparison of NN-based universal lossless compression methods

Method	Model Pre-training	Model Saving	Compression Ratio	Time Cost	Representative Works
Static Pre-training	✓	✓	⋆ ⋆ ⋆	⋆	Ref [78, 80, 81], DeepZip [82], DecMac [83], LLMZip [84]
Adaptive	✗	✗	⋆	⋆ ⋆ ⋆	TRACE [37],OREA [38], PAC [39], Cmix [85]
Semi-Adaptive	✓	✓	⋆⋆	⋆⋆	DZip (Combined Model) [79]

Notes. **Model Pre-training**: Whether or not the model $\mathcal{M}$ requires pre-training. **Model Saving**: Whether or not the model need to be saved and included in the compressed file. **Compression Ratio**: A smaller value indicates better compression performance, detailed in the evaluation metrics section. The " ✓ " and " ✗ " represent " Yes " or " No ", and more instances of " ⋆ " indicate a stronger advantage.